



Practica 1

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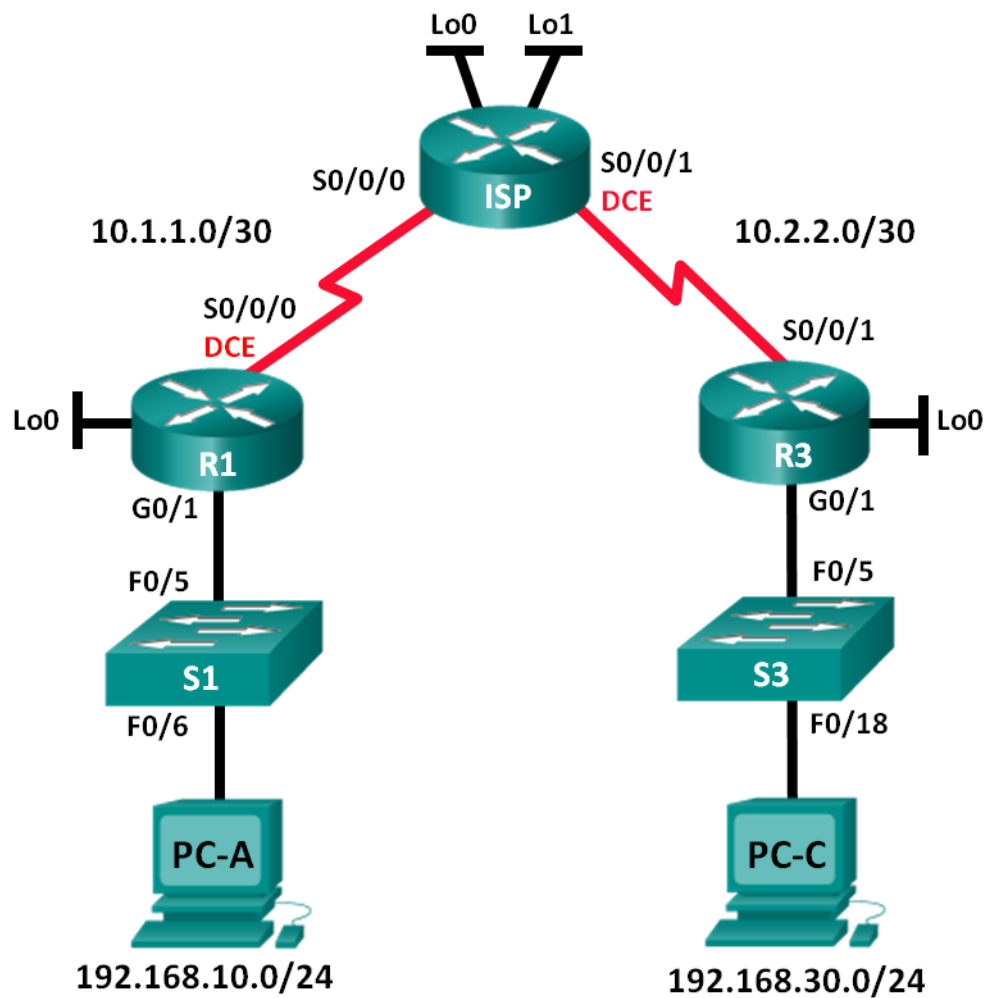
CIB-12 SEGURIDAD EN REDES

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Lab – Configuring and Verifying Extended ACLs

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/1	192.168.10.1	255.255.255.0	N/A
	Lo0	192.168.20.1	255.255.255.0	N/A
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A
ISP	S0/0/0	10.1.1.2	255.255.255.252	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A
	Lo0	209.165.200.225	255.255.255.224	N/A
	Lo1	209.165.201.1	255.255.255.224	N/A
R3	G0/1	192.168.30.1	255.255.255.0	N/A
	Lo0	192.168.40.1	255.255.255.0	N/A
	S0/0/1	10.2.2.1	255.255.255.252	N/A
S1	VLAN 1	192.168.10.11	255.255.255.0	192.168.10.1
S3	VLAN 1	192.168.30.11	255.255.255.0	192.168.30.1
PC-A	NIC	192.168.10.3	255.255.255.0	192.168.10.1
PC-C	NIC	192.168.30.3	255.255.255.0	192.168.30.1

Objectives

Part 1: Set Up the Topology and Initialize Devices

Part 2: Configure Devices and Verify Connectivity

- Configure basic settings on PCs, routers, and switches.
- Configure OSPF routing on R1, ISP, and R3.

Part 3: Configure and Verify Extended Numbered and Named ACLs

- Configure, apply, and verify a numbered extended ACL.
- Configure, apply, and verify a named extended ACL.

Part 4: Modify and Verify Extended ACLs

Background / Scenario

Extended access control lists (ACLs) are extremely powerful. They offer a much greater degree of control than standard ACLs as to the types of traffic that can be filtered, as well as where the traffic originated and where it is going.

In this lab, you will set up filtering rules for two offices represented by R1 and R3. Management has established some access policies between the LANs located at R1 and R3, which you must implement. The ISP router between R1 and R3 does not have any ACLs placed on it. You would not be allowed any administrative access to an ISP router as you can only control and manage your own equipment.

Note: The routers used with CCNA hands-on labs are Cisco 1941 Integrated Services Routers (ISRs) with Cisco IOS Release 15.2(4)M3 (universalk9 image). The switches used are Cisco Catalyst 2960s with Cisco IOS Release 15.0(2) (lanbasek9 image). Other routers, switches, and Cisco IOS versions can be used.

Depending on the model and Cisco IOS version, the commands available and output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Make sure that the routers and switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 3 Routers (Cisco 1941 with Cisco IOS Release 15.2(4)M3 universal image or comparable)
- 2 Switches (Cisco 2960 with Cisco IOS Release 15.0(2) lanbasek9 image or comparable)
- 2 PCs (Windows 7, Vista, or XP with terminal emulation program, such as Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet and serial cables as shown in the topology

Part 1: Set Up the Topology and Initialize Devices

In Part 1, you will set up the network topology and clear any configurations if necessary.

Step 1: Cable the network as shown in the topology.

Step 2: Initialize and reload the routers and switches.

Part 2: Configure Devices and Verify Connectivity

In Part 2, you will configure basic settings on the routers, switches, and PCs. Refer to the Topology and Addressing Table for device names and address information.

Step 1: Configure IP addresses on PC-A and PC-C.

Step 2: Configure basic settings on R1.

- Disable DNS lookup.
- Configure the device name as shown in the topology.
- Create a loopback interface on R1.
- Configure interface IP addresses as shown in the Topology and Addressing Table.
- Configure a privileged EXEC mode password of **class**.
- Assign a clock rate of **128000** to the S0/0/0 interface.
- Assign **cisco** as the console and vty password and enable Telnet access. Configure **logging synchronous** for both the console and vty lines.
- Enable web access on R1 to simulate a web server with local authentication for user **admin**.

```
R1(config)# ip http server
R1(config)# ip http authentication local
R1(config)# username admin privilege 15 secret class
```

Step 3: Configure basic settings on ISP.

- Configure the device name as shown in the topology.

- b. Create the loopback interfaces on ISP.
- c. Configure interface IP addresses as shown in the Topology and Addressing Table.
- d. Disable DNS lookup.
- e. Assign **class** as the privileged EXEC mode password.
- f. Assign a clock rate of **128000** to the S0/0/1 interface.
- g. Assign **cisco** as the console and vty password and enable Telnet access. Configure **logging synchronous** for both console and vty lines.
- h. Enable web access on the ISP. Use the same parameters as in Step 2h.

Step 4: Configure basic settings on R3.

- a. Configure the device name as shown in the topology.
- b. Create a loopback interface on R3.
- c. Configure interface IP addresses as shown in the Topology and Addressing Table.
- d. Disable DNS lookup.
- e. Assign **class** as the privileged EXEC mode password.
- f. Assign **cisco** as the console password and configure **logging synchronous** on the console line.
- g. Enable SSH on R3.

```
R3(config)# ip domain-name cisco.com
R3(config)# crypto key generate rsa modulus 1024
R3(config)# line vty 0 4
R3(config-line)# login local
R3(config-line)# transport input ssh
```

- h. Enable web access on R3. Use the same parameters as in Step 2h.

Step 5: (Optional) Configure basic settings on S1 and S3.

- a. Configure the hostnames as shown in the topology.
- b. Configure the management interface IP addresses as shown in the Topology and Addressing Table.
- c. Disable DNS lookup.
- d. Configure a privileged EXEC mode password of **class**.
- e. Configure a default gateway address.

Step 6: Configure OSPF routing on R1, ISP, and R3.

- a. Assign 1 as the OSPF process ID and advertise all networks on R1, ISP, and R3. The OSPF configuration for R1 is included for reference.

```
R1(config)# router ospf 1
R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
R1(config-router)# network 192.168.20.0 0.0.0.255 area 0
R1(config-router)# network 10.1.1.0 0.0.0.3 area 0
```

- b. After configuring OSPF on R1, ISP, and R3, verify that all routers have complete routing tables listing all networks. Troubleshoot if this is not the case.

Step 7: Verify connectivity between devices.

Note: It is very important to verify connectivity **before** you configure and apply ACLs! Ensure that your network is properly functioning before you start to filter out traffic.

- a. From PC-A, ping PC-C and the loopback and serial interfaces on R3.

```
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.3

Pinging 192.168.30.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.3: bytes=32 time=19ms TTL=125
Reply from 192.168.30.3: bytes=32 time=12ms TTL=125
Reply from 192.168.30.3: bytes=32 time=18ms TTL=125

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 19ms, Average = 16ms

C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time=14ms TTL=253
Reply from 192.168.40.1: bytes=32 time=8ms TTL=253
Reply from 192.168.40.1: bytes=32 time=11ms TTL=253
Reply from 192.168.40.1: bytes=32 time=13ms TTL=253

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 14ms, Average = 11ms

C:\>ping 10.2.2.1

Pinging 10.2.2.1 with 32 bytes of data:

Reply from 10.2.2.1: bytes=32 time=19ms TTL=253
Reply from 10.2.2.1: bytes=32 time=13ms TTL=253
Reply from 10.2.2.1: bytes=32 time=14ms TTL=253
Reply from 10.2.2.1: bytes=32 time=12ms TTL=253

Ping statistics for 10.2.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 19ms, Average = 14ms

C:\>
```

Were your pings successful? Yes

- b. From R1, ping PC-C and the loopback and serial interface on R3.

```
R1#ping 192.168.30.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.3, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/15/18 ms

R1#

R1#ping 192.168.40.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 11/13/15 ms

R1#
```

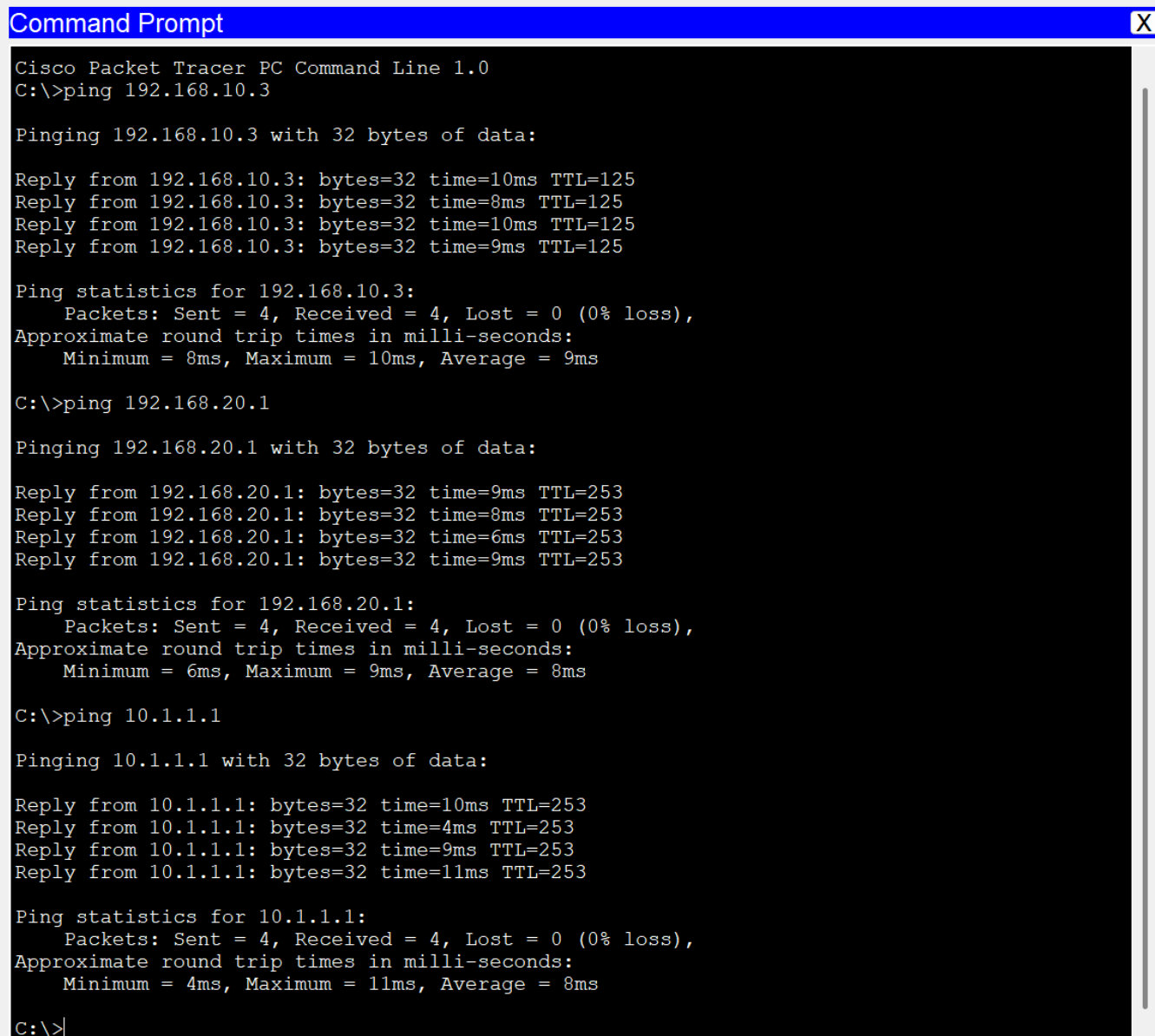
```
R1#ping 10.2.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 10/12/15 ms

R1#
```

Were your pings successful? Yes

- c. From PC-C, ping PC-A and the loopback and serial interface on R1.



```
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=10ms TTL=125
Reply from 192.168.10.3: bytes=32 time=8ms TTL=125
Reply from 192.168.10.3: bytes=32 time=10ms TTL=125
Reply from 192.168.10.3: bytes=32 time=9ms TTL=125

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 10ms, Average = 9ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=9ms TTL=253
Reply from 192.168.20.1: bytes=32 time=8ms TTL=253
Reply from 192.168.20.1: bytes=32 time=6ms TTL=253
Reply from 192.168.20.1: bytes=32 time=9ms TTL=253

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 9ms, Average = 8ms

C:\>ping 10.1.1.1

Pinging 10.1.1.1 with 32 bytes of data:

Reply from 10.1.1.1: bytes=32 time=10ms TTL=253
Reply from 10.1.1.1: bytes=32 time=4ms TTL=253
Reply from 10.1.1.1: bytes=32 time=9ms TTL=253
Reply from 10.1.1.1: bytes=32 time=11ms TTL=253

Ping statistics for 10.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 11ms, Average = 8ms

C:\>
```

Were your pings successful? Yes

- d. From R3, ping PC-A and the loopback and serial interface on R1.

```
R3#ping 192.168.10.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.3, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/13/22 ms

R3#ping 192.168.20.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/5 ms

R3#ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/10/14 ms

R3#
```

Were your pings successful? Yes

- e. From PC-A, ping the loopback interfaces on the ISP router.

```
Pinging 209.165.200.225 with 32 bytes of data:

Reply from 209.165.200.225: bytes=32 time=11ms TTL=254
Reply from 209.165.200.225: bytes=32 time=1ms TTL=254
Reply from 209.165.200.225: bytes=32 time=8ms TTL=254
Reply from 209.165.200.225: bytes=32 time=14ms TTL=254

Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 8ms

C:\>ping 209.165.201.1

Pinging 209.165.201.1 with 32 bytes of data:

Reply from 209.165.201.1: bytes=32 time=12ms TTL=254
Reply from 209.165.201.1: bytes=32 time=9ms TTL=254
Reply from 209.165.201.1: bytes=32 time=6ms TTL=254
Reply from 209.165.201.1: bytes=32 time=5ms TTL=254

Ping statistics for 209.165.201.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 12ms, Average = 8ms

C:\>
```

Were your pings successful? Yes

- f. From PC-C, ping the loopback interfaces on the ISP router.


```
Pinging 209.165.201.1 with 32 bytes of data:

Reply from 209.165.201.1: bytes=32 time=8ms TTL=254
Reply from 209.165.201.1: bytes=32 time=4ms TTL=254
Reply from 209.165.201.1: bytes=32 time=1ms TTL=254
Reply from 209.165.201.1: bytes=32 time=4ms TTL=254

Ping statistics for 209.165.201.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 8ms, Average = 4ms

C:\>ping 209.165.200.225

Pinging 209.165.200.225 with 32 bytes of data:

Reply from 209.165.200.225: bytes=32 time=2ms TTL=254
Reply from 209.165.200.225: bytes=32 time=2ms TTL=254
Reply from 209.165.200.225: bytes=32 time=2ms TTL=254
Reply from 209.165.200.225: bytes=32 time=1ms TTL=254

Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>
```

Were your pings successful? Yes

- g. Open a web browser on PC-A and go to <http://209.165.200.225> on ISP. You will be prompted for a username and password. Use **admin** for the username and **class** for the password. If you are prompted to accept a signature, accept it. The router will load the Cisco Configuration Professional (CCP) Express in a separate window. You may be prompted for a username and password. Use **admin** for the username and **class** for the password.
- h. Open a web browser on PC-C and go to <http://10.1.1.1> on R1. You will be prompted for a username and password. Use **admin** for username and **class** for the password. If you are prompted to accept a signature, accept it. The router will load CCP Express in a separate window. You may be prompted for a username and password. Use **admin** for the username and **class** for the password.

Part 3: Configure and Verify Extended Numbered and Named ACLs

Extended ACLs can filter traffic in many different ways. Extended ACLs can filter on source IP addresses, source ports, destination IP addresses, destination ports, as well as various protocols and services.

Security policies are as follows:

1. Allow web traffic originating from the 192.168.10.0/24 network to go to any network.
2. Allow an SSH connection to the R3 serial interface from PC-A.
3. Allow users on 192.168.10.0/24 network access to 192.168.20.0/24 network.

4. Allow web traffic originating from the 192.168.30.0/24 network to access R1 via the web interface and the 209.165.200.224/27 network on ISP. The 192.168.30.0/24 network should NOT be allowed to access any other network via the web.

In looking at the security policies listed above, you will need at least two ACLs to fulfill the security policies. A best practice is to place extended ACLs as close to the source as possible. We will follow this practice for these policies.

Step 1: Configure a numbered extended ACL on R1 for security policy numbers 1 and 2.

You will use a numbered extended ACL on R1. What are the ranges for extended ACLs?

-
- a. Configure the ACL on R1. Use 100 for the ACL number.

```
R1(config)# access-list 100 remark Allow Web & SSH Access
R1(config)# access-list 100 permit tcp host 192.168.10.3 host 10.2.2.1 eq 22
R1(config)# access-list 100 permit tcp any any eq 80
```

What does the 80 signify in the command output listed above?

El número 80 en el comando eq 80 especifica el puerto de destino para el tráfico HTTP, que es el puerto estándar para este protocolo.

To what interface should ACL 100 be applied?

La ACL 100 debe aplicarse a la interfaz S0/0/0 de R1, ya que es la interfaz de salida hacia la red de destino

In what direction should ACL 100 be applied?

La ACL 100 debe aplicarse en la dirección de salida (out) de la interfaz S0/0/0, para controlar el tráfico que sale de R1 hacia otras redes.

- b. Apply ACL 100 to the S0/0/0 interface.

```
R1(config)# interface s0/0/0
R1(config-if)# ip access-group 100 out
```

- c. Verify ACL 100.

- 1) Open up a web browser on PC-A, and access <http://209.165.200.225> (the ISP router). It should be successful; troubleshoot, if not.
- 2) Establish an SSH connection from PC-A to R3 using 10.2.2.1 for the IP address. Log in with **admin** and **class** for your credentials. It should be successful; troubleshoot, if not.
- 3) From privileged EXEC mode prompt on R1, issue the **show access-lists** command.

```
R1# show access-lists
Extended IP access list 100
 10 permit tcp host 192.168.10.3 host 10.2.2.1 eq 22 (22 matches)
 20 permit tcp any any eq www (111 matches)
```

```
R1#show access-lists
Extended IP access list 100
 10 permit tcp host 192.168.10.3 host 10.2.2.1 eq 22
 20 permit tcp any any eq www
R1#
```

- 4) From the PC-A command prompt, issue a ping to 10.2.2.1. Explain your results.

```
C:\>ping 10.2.2.1

Pinging 10.2.2.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 10.2.2.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```

En mi caso fallo lo que significa que puede ser necesario ajustar la ACL para permitir este tipo de tráfico.

Step 2: Configure a named extended ACL on R3 for security policy number 3.

- a. Configure the policy on R3. Name the ACL WEB-POLICY.

```
R3(config)# ip access-list extended WEB-POLICY
R3(config-ext-nacl)# permit tcp 192.168.30.0 0.0.0.255 host 10.1.1.1 eq 80
R3(config-ext-nacl)# permit tcp 192.168.30.0 0.0.0.255 209.165.200.224
0.0.0.31 eq 80
```

- b. Apply ACL WEB-POLICY to the S0/0/1 interface.

```
R3(config-ext-nacl)# interface S0/0/1
R3(config-if)# ip access-group WEB-POLICY out
```

- c. Verify the ACL WEB-POLICY.

- 1) From R3 privileged EXEC mode command prompt, issue the **show ip interface s0/0/1** command.

```
R3#show ip interface s0/0/1
Serial0/0/1 is up, line protocol is up (connected)
Internet address is 10.2.2.1/30
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is WEB-POLICY
Inbound access list is not set
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachable are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
WCCP Redirect outbound is disabled
WCCP Redirect exclude is disabled
BGP Policy Mapping is disabled

R3#
```

What, if any, is the name of the ACL? WEB-POLICY

In what direction is the ACL applied? Salida (out)

- 2) Open up a web browser on PC-C and access <http://209.165.200.225> (the ISP router). It should be successful; troubleshoot, if not.
Despues de utilizar el comando
- 3) From PC-C, open a web session to <http://10.1.1.1> (R1). It should be successful; troubleshoot, if not.
- 4) From PC-C, open a web session to <http://209.165.201.1> (ISP router). It should fail; troubleshoot, if not.
- 5) From a PC-C command prompt, ping PC-A. What was your result and why?

El ping Fallo ya que ACL bloquea tráfico entre las redes.

```
C:\> ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Part 4: Modify and Verify Extended ACLs

Because of the ACLs applied on R1 and R3, no pings or any other kind of traffic is allowed between the LAN networks on R1 and R3. Management has decided that all traffic between the 192.168.10.0/24 and 192.168.30.0/24 networks should be allowed. You must modify both ACLs on R1 and R3.

Step 1: Modify ACL 100 on R1.

- a. From R1 privileged EXEC mode, issue the **show access-lists** command.

```
R1>ena
Password:
R1#show access-lists
Extended IP access list 100
 10 permit tcp host 192.168.10.3 host 10.2.2.1 eq 22
 20 permit tcp any any eq www (1 match(es))
```

R1#

How many lines are there in this access list? 2

- b. Enter global configuration mode and modify the ACL on R1.

```
R1(config)# ip access-list extended 100
R1(config-ext-nacl)# 30 permit ip 192.168.10.0 0.0.0.255 192.168.30.0
0.0.0.255
R1(config-ext-nacl)# end
```

- c. Issue the **show access-lists** command.

Where did the new line that you just added appear in ACL 100?

```
R1#show access-lists
Extended IP access list 100
 10 permit tcp host 192.168.10.3 host 10.2.2.1 eq 22
 20 permit tcp any any eq www (1 match(es))
 30 permit ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
R1#
```

30 permit ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255

Step 2: Modify ACL WEB-POLICY on R3.

- a. From R3 privileged EXEC mode, issue the **show access-lists** command.

```
R3#show access-lists
Extended IP access list WEB-POLICY
 10 permit tcp 192.168.30.0 0.0.0.255 host 10.1.1.1 eq www
 20 permit tcp 192.168.30.0 0.0.0.255 209.165.200.224 0.0.0.31 eq www (2 match(es))
R3#
```

How many lines are there in this access list? 2

- b. Enter global configuration mode and modify the ACL on R3.

```
R3(config)# ip access-list extended WEB-POLICY
R3(config-ext-nacl)# 30 permit ip 192.168.30.0 0.0.0.255 192.168.10.0
0.0.0.255
R3(config-ext-nacl)# end
```

- c. Issue the **show access-lists** command to verify that the new line was added at the end of the ACL.

```
R3#show access-lists
Extended IP access list WEB-POLICY
 10 permit tcp 192.168.30.0 0.0.0.255 host 10.1.1.1 eq www
 20 permit tcp 192.168.30.0 0.0.0.255 209.165.200.224 0.0.0.31 eq www (2 match(es))
 30 permit ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
R3#
```

Step 3: Verify modified ACLs.

- a. From PC-A, ping the IP address of PC-C. Were the pings successful? Si

```
C:\>ping 192.168.30.3

Pinging 192.168.30.3 with 32 bytes of data:

Reply from 192.168.30.3: bytes=32 time=14ms TTL=125
Reply from 192.168.30.3: bytes=32 time=12ms TTL=125
Reply from 192.168.30.3: bytes=32 time=9ms TTL=125
Reply from 192.168.30.3: bytes=32 time=14ms TTL=125

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 14ms, Average = 12ms

C:\>
```

- b. From PC-C, ping the IP address of PC-A. Were the pings successful? Si

```
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=18ms TTL=125
Reply from 192.168.10.3: bytes=32 time=9ms TTL=125
Reply from 192.168.10.3: bytes=32 time=2ms TTL=125
Reply from 192.168.10.3: bytes=32 time=12ms TTL=125

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 18ms, Average = 10ms

C:\>
```

Why did the ACLs work immediately for the pings after you changed them?

Porque las ACLs se aplican de inmediato al tráfico; no es necesario reiniciar interfaces o routers. Además, el protocolo ICMP (usado por ping) está permitido explícitamente en las reglas nuevas.

Reflection

1. Why is careful planning and testing of ACLs required?

- Porque una ACL mal configurada puede bloquear tráfico legítimo o permitir tráfico no deseado, afectando la seguridad y el funcionamiento de la red.
- Además, una ACL errónea puede causar interrupciones en los servicios y dificultar la resolución de problemas.

2. Which type of ACL is better: standard or extended?

- Las **ACL extendidas** son mejores porque permiten filtrar por múltiples criterios, como origen, destino, protocolos y puertos, ofreciendo mayor control y seguridad.
- Las ACL estándar solo filtran por dirección IP de origen, lo que limita su uso en entornos con necesidades más específicas.
- Sin embargo, las ACL estándar son más simples y pueden ser útiles para filtros básicos.

3. Why are OSPF hello packets and routing updates not blocked by the implicit **deny any** access control entry (ACE) or ACL statement of the ACLs applied to R1 and R3?

- Porque OSPF utiliza protocolos de enrutamiento que generalmente usan protocolos IP específicos (como protocolo 89 para OSPF) y, en muchos casos, las ACLs se configuran para permitir tráfico de enrutamiento o están aplicadas solo en interfaces específicas que no afectan esos paquetes.
- Además, algunas implementaciones de Cisco permiten tráfico de protocolos de enrutamiento antes de aplicar ACLs o a través de rutas administrativas que no se ven afectadas.
- También puede ser porque la ACL no filtra tráfico del protocolo OSPF específicamente, y por defecto no bloquea tráfico relacionado con protocolos críticos de enrutamiento.

Router Interface Summary Table

Router Interface Summary				
Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
1800	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
1900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2801	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
2811	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. There is no way to effectively list all the combinations of configurations for each router class. This table includes identifiers for the possible combinations of Ethernet and Serial interfaces in the device. The table does not include any other type of interface, even though a specific router may contain one. An example of this might be an ISDN BRI interface. The string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface.				